

Platform Example

Immunization Write Path — V2 FHIRconnect

Full FHIR Bundle → COMPOSITION-level openFHIR mapping → openEHR → EHRbase, with ~20 mapped fields.

- Platform example (Kotlin + Ktor) — orchestrates the stack
- fhir-server-1 (HAPI 8.8.1) — demographics + directory store (Patient / Practitioner / Organization)
- openFHIR 2.2.3 — bidirectional FHIR ↔ openEHR mapping engine (FHIRconnect V2 YAMLS)
- EHRbase 2.31.0 — openEHR Composition store + AQL

V2 mapping: COMPOSITION-level, ~20 ACTION fields, multi-immunization Bundles, vaccination protocol, adverse reactions, performer, verification status, feeder audit.

Bidirectional mapping loaded (admin + vaccination-record); write path tested end-to-end.

The Write Flow (V2)

POST /Immunization — full Bundle in, COMPOSITION-level mapping, multi-immunization support

- 1. Client**
POSTs a CH VACD Immunization Administration Document Bundle
to platform /Immunization
- 2. Demographics fan-out**
Patient + Practitioner + Organization → fhir-server-1 (HAPI)
Platform captures the HAPI-assigned ids
- 3. EHR resolution**
Find-or-create openEHR EHR in EHRbase
Linked via EHR_STATUS.subject.external_ref (namespace=ch-vacd, id=<Patient.id>)
- 4. V2 Conversion**
Full Bundle → openFHIR /openfhir/toopenehr (COMPOSITION-level mapping)
openFHIR resolves urn:uuid refs, navigates sections, maps ~20 fields per Immunization
- 5. Enrich**
Platform adds ctx/ metadata (language, territory, composer) for EHRbase
feederAudit handled by V2 mapping — putIfAbsent fallback for Konkretisierung §13
- 6. Persist**
POST /ehr/{ehrId}/composition?format=FLAT&templateId=...
Returns 201 + compositionUid; one Composition may contain multiple ACTION entries

Demo UI — initial state

/demo loads with the first example pre-populated.

● CH VACDPlatform Business Logic Example (harness smoke-test)

healthzmetadataCH VACD IG ↗

Example: CH VACD Immunization Administration (f79a544f) ▾

Convert & Store ▶

1 · FHIR Immunization (input)

application/fhir+json

```
{
  "resourceType": "Bundle",
  "id": "3c12e373-0c33-4fa1-9067-ca7de785477d",
  "meta": {
    "profile": [
      "http://fhir.ch/ig/ch-vacd/StructureDefinition/ch-vacd-document-immunization-administration"
    ]
  },
  "language": "en-US",
  "identifier": {
    "system": "urn:ietf:rfc:3986",
    "value": "urn:uuid:3c12e373-0c33-4fa1-9067-ca7de785477d"
  },
  "type": "document",
  "timestamp": "2026-05-23T21:31:58.228+02:00",
  "entry": [
    {
      "fullUrl": "urn:uuid:fb68fc23-a151-4ea0-a575-a9a59a8b10eb",
      "resource": {
        "resourceType": "Composition",
        "meta": {
```

2 · openEHR FLAT Composition (after openFHIR)

intermediate

FLAT JSON will appear here.

3 · EHRbase confirmation

AQL roundtrip

Composition UID, EHR ID, AQL result.

How this works

1. Client POSTs a CH VACD Immunization Administration Document Bundle to `POST /Immunization`.
2. The platform fans Patient/Practitioner/Organization to `fhir-server-1` (HAPI). The FHIR server is the demographics + directory store.
3. The platform ensures an openEHR EHR exists in `EHRbase`, linked to the Patient via `EHR_STATUS.subject.external_ref`.
4. The platform sends the full Bundle to `openFHIR /openfhir/toopenehr` — the V2 FHIRconnect mapping converts it to a FLAT openEHR Composition (COMPOSITION-level mapping with ~20 fields).
5. The platform enriches FLAT with `ctx/` metadata (language, territory, composer) that EHRbase requires.
6. The platform persists the Composition to EHRbase and returns `201 Created` with `compositionUid`.

Hackathon harness — services: HAPI 8.8.1 · openFHIR 2.2.3 · EHRbase 2.31.0

screenshot: 00-initial.png

Mixed vaccines — Boostrix Polio + FSME-Immun

3 immunizations: 2× Boostrix Polio (681) + 1× FSME-Immun Junior (683). Patient: Monika Wegmueller.

● CH VACD Platform Business Logic Example (harness smoke-test)

healthz metadata CH VACD IG ↗

Example: CH VACD Immunization Administration (f79a544f) ▾

Convert & Store ▶

1 · FHIR Immunization (input)

application/fhir+json

```
{
  "resourceType": "Bundle",
  "id": "3c12e373-0c33-4fa1-9067-ca7de785477d",
  "meta": {
    "profile": [
      "http://fhir.ch/ig/ch-vacd/StructureDefinition/ch-vacd-document-immunization-administration"
    ]
  },
  "language": "en-US",
  "identifier": {
    "system": "urn:ietf:rfc:3986",
    "value": "urn:uuid:3c12e373-0c33-4fa1-9067-ca7de785477d"
  },
  "type": "document",
  "timestamp": "2026-05-23T21:31:58.228+02:00",
  "entry": [
    {
      "fullUrl": "urn:uuid:fb68fc23-a151-4ea0-a575-a9a59a8b10eb",
      "resource": {
        "resourceType": "Composition",
        "meta": {
          "profile": [
            "http://fhir.ch/ig/ch-
```

2 · openEHR FLAT Composition (after openFHIR)

intermediate

```
{
  "ch_vacd_immunization_administration/medication_management:0/time": "2009-04-22T00:00:00",
  "ch_vacd_immunization_administration/medication_management:0/ism_transition/careflow_step|terminology": "local",
  "ch_vacd_immunization_administration/medication_management:0/ism_transition/careflow_step|code": "at0006",
  "ch_vacd_immunization_administration/medication_management:0/ism_transition/careflow_step|value": "Dose administration",
  "ch_vacd_immunization_administration/medication_management:0/other_participation:0|id": "urn:uuid:0faa55b6-0cc2-44ea-a74e-cc8b2b4ae341",
  "ch_vacd_immunization_administration/medication_management:0/other_participation:0|id_namespace": "http://hl7.org/fhir",
  "ch_vacd_immunization_administration/medication_management:0/vaccine_administered|code": "681",
  "ch_vacd_immunization_administration/medication_management:0/vaccine_administered|terminology": "http://fhir.ch/ig/ch-vacd/CodeSystem/ch-vacd-swissmedic-cs",
  "ch_vacd_immunization_administration/medication_management:0/vaccine_administered|value": "Boostrix
```

3 · EHRbase confirmation

AQL roundtrip

compositionUid	0ec84ad2-be33-47fe-a259-98ee7692501e::local.ehrbase.org::1
ehrId	765f8025-93ff-4527-8dfa-bb296d0727f9
patientId (HAPI)	6ddc4cff-c6cd-4227-9f48-43bcdbe95688
practitionerIds	b669b9fe-ccd4-4daa-ba4f-6abc0a6dfa37, 75ba4030-4ac5-4b40-bafe-e76a56886fce, e828e437-b321-44e7-a7f9-083ca6037a4d
organizationIds	3ba429c5-c8b6-491e-a352-b4322ac2f8f4, 6e1b1550-c8fa-47b9-89e5-097356ae86986, 20917077-343b-488f-80d3-e070c2453041

AQL roundtrip

```
{
  "meta": {
    "type": "RESULTSET",
    "schema version": "1.0.3",
    "created": "2026-05-27T14:58:16.320812304Z",
    "executed_aql": "SELECT c/uid/value AS uid, c/archetype_details/template_id/value AS template FROM EHR e[ehr_id/value='765f8025-93ff-4527-8dfa-bb296d0727f9'] CONTAINS COMPOSITION c, \"resultsize\":1,\"q\":\"SELECT c/uid/value AS uid, c/archetype_details/template_id/value AS template FROM EHR e CONTAINS COMPOSITION c WHERE e/ehr_id/value = '765f8025-93ff-4527-8dfa-bb296d0727f9', \"columns\": [{\"path\":\"c/uid/value\", \"name\":\"uid\"}, {\"path\":\"c/archetype_details/template_id/value\", \"name\":\"templat
```

How this works

- Client POSTs a CH VACD Immunization Administration Document Bundle to `POST /Immunization`.
- The platform fans Patient/Practitioner/Organization to `fhir-server-1` (HAPI). The FHIR server is the demographics + directory store.
- The platform ensures an openEHR EHR exists in `EHRbase`, linked to the Patient via `EHR_STATUS.subject.external_ref`.
- The platform sends the full Bundle to `openFHIR /openfhir/toopenehr` — the V2 FHIRconnect mapping converts it to a FLAT openEHR Composition (COMPOSITION-level mapping with ~20 fields).
- The platform enriches FLAT with `ctx/` metadata (language, territory, composer) that EHRbase requires.
- The platform persists the Composition to EHRbase and returns `201 Created` with `compositionUid`.

Hackathon harness — services: HAPI 8.8.1 · openFHIR 2.2.3 · EHRbase 2.31.0

screenshot: 02-chvacd-immunizationadministration-6b3a054b-8cb4-44b4-bde8-f79a544f5b00-done.png

2-dose Comirnaty series

2 immunizations: Comirnaty COVID-19 (68225). Patient: Max Muster.

● CH VACD Platform Business Logic Example (harness smoke-test)

healthz metadata CH VACD IG ↗

Example: CH VACD Immunization Administration (3634ac7b) ▾

Convert & Store ▶

1 · FHIR Immunization (input)

application/fhir+json

```
{
  "resourceType": "Bundle",
  "id": "20f16d0b-fdc9-423f-a3b8-38e9f45247f0",
  "meta": {
    "profile": [
      "http://fhir.ch/ig/ch-vacd/StructureDefinition/ch-vacd-document-immunization-administration"
    ]
  },
  "language": "en-US",
  "identifier": {
    "system": "urn:ietf:rfc:3986",
    "value": "urn:uuid:20f16d0b-fdc9-423f-a3b8-38e9f45247f0"
  },
  "type": "document",
  "timestamp": "2026-05-23T21:21:17.397+02:00",
  "entry": [
    {
      "fullUrl": "urn:uuid:11ddeaab-1cd5-40b9-a973-82b0bf21612c",
      "resource": {
        "resourceType": "Composition",
        "meta": {
          "profile": [
            "http://fhir.ch/ig/ch-
```

2 · openEHR FLAT Composition (after openFHIR)

intermediate

```
{
  "ch_vacd_immunization_administration/medication_management:0/time": "2021-05-22T00:00:00",
  "ch_vacd_immunization_administration/medication_management:0/ism_transition/careflow_step|terminology": "local",
  "ch_vacd_immunization_administration/medication_management:0/ism_transition/careflow_step|code": "at0006",
  "ch_vacd_immunization_administration/medication_management:0/ism_transition/careflow_step|value": "Dose administration",
  "ch_vacd_immunization_administration/medication_management:0/other_participation:0|id": "urn:uid:80945811-f4de-454b-b868-94b577affc34",
  "ch_vacd_immunization_administration/medication_management:0/other_participation:0|id_namespace": "http://hl7.org/fhir",
  "ch_vacd_immunization_administration/medication_management:0/vaccine_administered|code": "68225",
  "ch_vacd_immunization_administration/medication_management:0/vaccine_administered|terminology": "http://fhir.ch/ig/ch-vacd/CodeSystem/ch-vacd-swissmedic-cs",
  "ch_vacd_immunization_administration/medication_management:0/vaccine_administered|value": "Comirna
```

3 · EHRbase confirmation

AQL roundtrip

compositionUid	25c38228-e359-4ad7-8d15-46b7d794ce88::local.ehrbase.org::1
ehrId	15f39c85-859c-4d03-977d-d83f151987c4
patientId (HAPI)	8e9679cd-ee16-4cba-9d05-c2b85dc8809b
practitionerIds	2c4a5004-bbdd-4e05-b713-5dd331da3fa4, bf95fcfe-fd12-4115-a310-6d716084a5d5, 4706715f-c3bd-4e63-afea-40fd10327fc3
organizationIds	3d5841ae-ffb8-4f0a-af68-09b6dae45ed3, d7fe7e5e-8148-4d3c-8fe6-1b7815de80d1, a7a6b005-373b-4e1f-95de-26e440a1bd0e

AQL roundtrip

```
{
  "meta": {
    "type": "RESULTSET",
    "schema version": "1.0.3",
    "created": "2026-05-27T14:58:17.064925387Z",
    "executed_aql": "SELECT c/uid/value AS uid, c/archetype_details/template_id/value AS template FROM EHR e[ehr_id/value='15f39c85-859c-4d03-977d-d83f151987c4'] CONTAINS COMPOSITION c, \"resultsize\":1}, \"q\": \"SELECT c/uid/value AS uid, c/archetype_details/template_id/value AS template FROM EHR e CONTAINS COMPOSITION c WHERE e/ehr_id/value = '15f39c85-859c-4d03-977d-d83f151987c4', \"columns\": [{\"path\": \"c/uid/value\", \"name\": \"uid\"}, {\"path\": \"c/archetype_details/template_id/value\", \"name\": \"templat
```

How this works

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Hackathon harness — services: HAPI 8.8.1 · openFHIR 2.2.3 · EHRbase 2.31.0

screenshot: 04-chvacd-immunizationadministration-a16565ea-fdde-495a-9e3b-3634ac7bb304-done.png

2-dose Comirnaty series (variant)

2 immunizations: Comirnaty COVID-19 (68225). Patient: Max Muster.

● CH VACD Platform Business Logic Example (harness smoke-test)

healthz metadata CH VACD IG ↗

Example: CH VACD Immunization Administration (34390a08) ▾

Convert & Store ▶

1 · FHIR Immunization (input)

application/fhir+json

```
{
  "resourceType": "Bundle",
  "id": "19f519e9-83ea-46c1-a600-0569c416992a",
  "meta": {
    "profile": [
      "http://fhir.ch/ig/ch-vacd/StructureDefinition/ch-vacd-document-immunization-administration"
    ]
  },
  "language": "en-US",
  "identifier": {
    "system": "urn:ietf:rfc:3986",
    "value": "urn:uuid:19f519e9-83ea-46c1-a600-0569c416992a"
  },
  "type": "document",
  "timestamp": "2026-05-23T21:24:43.095+02:00",
  "entry": [
    {
      "fullUrl": "urn:uuid:530f6b74-44d9-442c-8218-b27e0a41d62b",
      "resource": {
        "resourceType": "Composition",
        "meta": {
          "profile": [
            "http://fhir.ch/ig/ch-
```

2 · openEHR FLAT Composition (after openFHIR)

intermediate

```
{
  "ch_vacd_immunization_administration/medication_management:0/time": "2021-05-22T00:00:00",
  "ch_vacd_immunization_administration/medication_management:0/ism_transition/careflow_step|terminology": "local",
  "ch_vacd_immunization_administration/medication_management:0/ism_transition/careflow_step|code": "at0006",
  "ch_vacd_immunization_administration/medication_management:0/ism_transition/careflow_step|value": "Dose administration",
  "ch_vacd_immunization_administration/medication_management:0/other_participation:0|id": "urn:uuid:8840b388-44b1-4dd9-aca4-6a423ed73e99",
  "ch_vacd_immunization_administration/medication_management:0/other_participation:0|id_namespace": "http://hl7.org/fhir",
  "ch_vacd_immunization_administration/medication_management:0/vaccine_administered|code": "68225",
  "ch_vacd_immunization_administration/medication_management:0/vaccine_administered|terminology": "http://fhir.ch/ig/ch-vacd/CodeSystem/ch-vacd-swissmedic-cs",
  "ch_vacd_immunization_administration/medication_management:0/vaccine_administered|value": "Comirna
```

3 · EHRbase confirmation

AQL roundtrip

compositionUid	59f90b64-9502-4e8c-b20a-34aab35eb60d::local.ehrbase.org::1
ehrId	70a35fd4-9c89-4cf8-a1e9-9d99704c2d7c
patientId (HAPI)	d1de7835-0e48-4551-baef-9d5a94b74b40
practitionerIds	f97f409c-2a73-4f35-9838-ac6231d23689, 8d2fc7b3-b691-40da-a2a3-bf1309a1a32b, 39420455-88d5-499a-bbda-e635cb11b7c6
organizationIds	9ea6b998-1813-435b-bb86-991d92f86197, a4b01501-91e0-4cf6-8ca3-c8dabbc90b02, 6705bd9d-e54d-4c96-976a-10dc64b5ffee

AQL roundtrip

```
{
  "meta": {
    "type": "RESULTSET",
    "schema_version": "1.0.3",
    "created": "2026-05-27T14:58:17.685500429Z",
    "executed_aql": "SELECT c/uid/value AS uid, c/archetype_details/template_id/value AS template FROM EHR e[ehr_id/value='70a35fd4-9c89-4cf8-a1e9-9d99704c2d7c'] CONTAINS COMPOSITION c, \"resultsize\":1}, \"q\": \"SELECT c/uid/value AS uid, c/archetype_details/template_id/value AS template FROM EHR e CONTAINS COMPOSITION c WHERE e/ehr_id/value = '70a35fd4-9c89-4cf8-a1e9-9d99704c2d7c', \"columns\": [{\"path\": \"c/uid/value\", \"name\": \"uid\"}, {\"path\": \"c/archetype_details/template_id/value\", \"name\": \"templat
```

How this works

- Client POSTs a CH VACD Immunization Administration Document Bundle to `POST /Immunization`.
- The platform fans Patient/Practitioner/Organization to `fhir-server-1` (HAPI). The FHIR server is the demographics + directory store.
- The platform ensures an openEHR EHR exists in `EHRbase`, linked to the Patient via `EHR_STATUS.subject.external_ref`.
- The platform sends the full Bundle to `openFHIR /openfhir/toopenehr` — the V2 FHIRconnect mapping converts it to a FLAT openEHR Composition (COMPOSITION-level mapping with ~20 fields).
- The platform enriches FLAT with `ctx/` metadata (language, territory, composer) that EHRbase requires.
- The platform persists the Composition to EHRbase and returns `201 Created` with `compositionUid`.

Hackathon harness — services: HAPI 8.8.1 · openFHIR 2.2.3 · EHRbase 2.31.0

screenshot: 06-chvacd-immunizationadministration-b6d3e7c8-58ad-4d17-b375-34390a08faec-done.png

3-dose Comirnaty 30µg series

3 immunizations: Comirnaty 30µg (68710-01). Patient: Max Muster.

● CH VACD Platform Business Logic Example (harness smoke-test)

healthz metadata CH VACD IG ↗

Example: CH VACD Immunization Administration (a638f7d7) ▾

Convert & Store ▶

1 · FHIR Immunization (input)

application/fhir+json

```
{
  "resourceType": "Bundle",
  "id": "cf36b4ea-feac-47d6-ab5c-99f35a88733c",
  "meta": {
    "profile": [
      "http://fhir.ch/ig/ch-vacd/StructureDefinition/ch-vacd-document-immunization-administration"
    ]
  },
  "language": "en-US",
  "identifier": {
    "system": "urn:ietf:rfc:3986",
    "value": "urn:uuid:cf36b4ea-feac-47d6-ab5c-99f35a88733c"
  },
  "type": "document",
  "timestamp": "2025-09-18T13:37:33.094+02:00",
  "entry": [
    {
      "fullUrl": "urn:uuid:a515fdc9-f019-4376-9ba8-5d172dc3be81",
      "resource": {
        "resourceType": "Composition",
        "meta": {
          "profile": [
            "http://fhir.ch/ig/ch-vacd/StructureDefinition/ch-vacd-composition-immunization-administration"
          ]
        }
      }
    }
  ]
}
```

2 · openEHR FLAT Composition (after openFHIR)

intermediate

```
{
  "ch_vacd_immunization_administration/medication_management:0/time": "2021-04-17T00:00:00",
  "ch_vacd_immunization_administration/medication_management:0/ism_transition/careflow_step|terminology": "local",
  "ch_vacd_immunization_administration/medication_management:0/ism_transition/careflow_step|code": "at0006",
  "ch_vacd_immunization_administration/medication_management:0/ism_transition/careflow_step|value": "Dose administration",
  "ch_vacd_immunization_administration/medication_management:0/other_participation:0|id": "urn:uuid:637385aa-88e3-4d90-aed6-cd01f1306a02",
  "ch_vacd_immunization_administration/medication_management:0/other_participation:0|id_namespace": "http://hl7.org/fhir",
  "ch_vacd_immunization_administration/medication_management:0/vaccine_administered|code": "68710-01",
  "ch_vacd_immunization_administration/medication_management:0/vaccine_administered|terminology": "http://fhir.ch/ig/ch-vacd/CodeSystem/ch-vacd-swissmedic-cs",
  "ch_vacd_immunization_administration/medication"
}
```

3 · EHRbase confirmation

AQL roundtrip

compositionUid	15ff1cb1-e9b6-4fd8-84a0-0f6ad0478e93::local.ehrbase.org::1
ehrId	98dbc786-6f7c-477c-96d0-533d115cf011
patientId (HAPI)	8657ea70-26e8-4d30-a18d-1ef40451ddef
practitionerIds	51ff4031-1193-4b87-a7dc-d3242cf3468f, f74e279a-ffa4-4f2b-92e4-530540eda47b, 06c939b2-7489-4dfe-b248-fa0793733b2a, 22eaa6e4-705f-47d7-98bb-9c303b249c08
organizationIds	c324c0d4-43eb-4fa3-8430-e2000e68ead3, 4ede0cde-e575-4f00-807e-e58246d12776, e3f283af-66eb-4081-b720-419f3628d1d1, afe2f781-fa85-4e8a-9c19-7f4c0485bad7

AQL roundtrip

```
{
  "meta": {
    "type": "RESULTSET",
    "schema_version": "1.0.3",
    "created": "2026-05-27T14:50:10.54870043Z",
    "executed_aql": "SELECT c/uid/value AS uid, c/archetype_detail_s/template_id/value AS template FROM EHR elehr_id/value = '98dbc786-6f7c-477c-96d0-533d115cf011' CONTAINS COMPOSITION c, 'resultsSize': 1, 'q': 'SELECT c/uid/value AS uid, c/archetype_details/template_id/value AS template FROM EHR e CONTAINS COMPOSITION c WHERE e/ehr_id/value = '98dbc786-6f7c-477c-96d0-533d115cf011', 'columns': [{ 'path': 'c/uid/value', 'name': 'uid' }, { 'path': 'c/archetype_details/template_id/value', 'name': 'template' } ], 'ro"
  }
}
```

How this works

1. Client POSTs a CH VACD Immunization Administration Document Bundle to `POST /Immunization`.
2. The platform fans Patient/Practitioner/Organization to `fhir-server-1` (HAPI). The FHIR server is the demographics + directory store.
3. The platform ensures an openEHR EHR exists in `EHRbase`, linked to the Patient via `EHR_STATUS.subject.external_ref`.
4. The platform sends the full Bundle to `openFHIR /openfhir/toopenehr` — the V2 FHIRconnect mapping converts it to a FLAT openEHR Composition (COMPOSITION-level mapping with ~20 fields).
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Hackathon harness — services: HAPI 8.8.1 · openFHIR 2.2.3 · EHRbase 2.31.0

screenshot: 08-chvacd-immunizationadministration-ec1548de-55c5-4081-9a96-a638f7d78a77-done.png

What's Proven

All validated end-to-end against the live stack with integration tests + AQL queries.

Proven (all green)

- ✓ V2 FHIRconnect mapping loaded as bidirectional unit (2 OPTs, 2 contexts, 4 models)
- ✓ openFHIR 2.2.3 COMPOSITION-level mapping: full Bundle in, ~20 fields per Immunization
- ✓ Multi-immunization Bundles (2-dose, 3-dose series) → single Composition, multiple ACTIONS
- ✓ Mixed vaccines (Boostrix Polio + FSME-Immun) in one Composition — different codes per ACTION
- ✓ AQL roundtrip: vaccineCode, time, ISM transition, doseNumber verified via queries
- ✓ feeder_audit: V2 mapping embeds original FHIR at COMPOSITION + ACTION level
- ✓ Performer participation: openFHIR maps actor ref, platform fills required RM fields
- ✓ V1 single-immunization Bundles still work (backward compatible)
- ✓ Demographics (Patient/Practitioner/Organization) landed in HAPI from document Bundles
- ✓ 12/12 tests pass (8 unit + 4 integration)

Next steps

- Read path — vaccination-record template + tofhir mapping (loaded, not yet tested)
- Profile validation — HAPI FhirValidator + CH VACD StructureDefinitions
- Terminology lookups — live tx.fhir.ch/r4 for code system validation